

Tran Minh Duong

AI Researcher & Applied AI Engineer

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About Me

AI Researcher & Applied AI Engineer specializing in Parameter-Efficient Fine-Tuning (PEFT/QLoRA), compact Vision-Language Models (VLMs), agentic workflows, and rigorous model evaluation.

Bridging empirical research with scalable production systems, with first-author/co-authored publications in Springer LNCS and Nature Portfolio.

Education

- BSc**

La Rochelle University, Informatique

La Rochelle, France

- **Valedictorian** (Ranked 1st in cohort) | GPA: 17.05/20 (≈ 3.89/4)
 - Thesis: "Historical Fidelity in OCR Post-Correction: A Comparative Study of Parameters Efficient Fine-Tuned LLMs and Seq2Seq Models"

Sept 2025 – June 2026
- BSc**

University of Science and Technology of Hanoi, ICT (Double Degree)

Hanoi, Vietnam

- **Valedictorian** (Ranked 1st of entire graduating class) | GPA: 18.48/20 (≈ 3.89/4)
 - Awarded 100% Merit Scholarship for 2 consecutive years

Sept 2023 – June 2026

Publications

- Metadata as Prompt Context for Historical OCR Post-Correction**

Dec 2026

Minh Duong Tran, Emanuela Boros, Antoine Doucet
(Proceedings of the 28th International Conference on Asia-Pacific Digital Libraries (ICADL 2026), Springer LNCS (Accepted, Oral Presentation Dec 2026))
- Towards Efficient Context-Aware Classification with Compact VLM Architectures: Indoor Fire Case Study**

Apr 2026

Anh Tuan Giang, Nhat Quang Doan, **Minh Duong Tran**, et al.
<https://doi.org/10.1038/s41598-026-48743-5> (Scientific Reports (Nature Portfolio, 2026))

Skills

- Programming Languages:** Python (proficient), Java, C++, Bash, SQL
- Framework / Libraries:** PyTorch, Hugging Face transformers, PEFT, bitsandbytes, vLLM, LangChain, FastAPI
- Databases & Vector Stores:** Neo4j (Graph), Pinecone (Vector), ChromaDB, PostgreSQL, Supabase, MongoDB
- Tools & DevOps:** AWS, Docker, Terraform, Git, GitHub Actions, Prometheus, Grafana, DeepEval
- Foreign Languages:** English C1 (IELTS 7.5), French B2 (DELF B2)

Experience

- FPT Software Innovation**, Junior AI Engineer

Hanoi, Vietnam

Develop enterprise AI systems, focusing on multi-agent architectures, context engineering, and production RAG pipelines.

June 2026 – present

- Built an end-to-end internal developer tool utilizing an **agentic workflow** with an **orchestrator/worker architecture**, context engineering, and reusable artifacts to accelerate legacy codebase migration
 - Designed enterprise RAG pipelines featuring **hybrid search** (dense + sparse) fused via **Reciprocal Rank Fusion (RRF)** and cross-encoder re-ranking for precise knowledge retrieval
 - Implemented multi-tool calling agents with structured execution schemas and **human-in-the-loop** verification gates
 - Hardened production AI services against **prompt injection** vulnerabilities, establishing token lifecycle management and API security protocols
 - Contributed **50+ modular AI components** (Skills, custom MCP servers, and agentic workflows) on internal plugin marketplace, recognized across innovation initiatives

L3i - La Rochelle Université, AI Researcher

La Rochelle, France

Conducted research on OCR Post-Correction for historical documents using modern LLMs.

Mar 2026 – June 2026

- Investigated decoder-only and seq2seq LLM architectures for historical document OCR post-correction
- Optimized Qwen3 and BART via **PEFT/QLoRA**, reducing OCR error rates by **up to 50%**
- Formulated **metadata-as-context conditioning** methodology and constructed qualitative error taxonomy for historical preservation failures

ICTLab - USTH, AI Research Assistant

Hanoi, Vietnam

Contributed to research on context-aware fire detection using Visual Language Models.

Sept 2024 – Aug 2025

- Co-authored paper published in **Scientific Reports (Nature Portfolio)** and released the [fire context awareness dataset](#)
- Evaluated compact VLMs (Qwen2.5-VL, InternVL) across custom metrics and inference latency under constrained edge compute
- Fine-tuned VLMs using **PEFT/LoRA** and quantization, boosting accuracy from **60% to 85%–90%**, tracked via Weights & Biases

Projects

HIPE-OCRepair [↗](#)

Accepted Research Codebase (ICADL 2026, Springer LNCS)

- Benchmarked decoder-only (Qwen3-7B/14B) and seq2seq (BART/mBART) across 100k+ historical lines in French, German, and English on the multilingual HIPE-OCRepair benchmark
- Formulated metadata-as-context conditioning (document date, type, domain), achieving up to **50% relative reduction in Character Error Rate (CER)** and **Word Error Rate (WER)** over vanilla OCR baselines
- Formulated a 5-class qualitative taxonomy to analyze hallucinations, entity drift, and historical syntax retention
- **Techstack:** PyTorch, Hugging Face Transformers, PEFT/QLoRA, bitsandbytes (4-bit NF4), Weights & Biases

ContextFire-VLM [↗](#)

Accepted Research Codebase (Scientific Reports, Nature Portfolio 2026)

- Curated and open-sourced a multimodal dataset of 10k+ labeled frames to benchmark compact VLMs (Qwen2.5-VL-3B/7B, InternVL2-2B/4B) distinguishing hazardous fires from safe heat sources
- Fine-tuned architectures using **PEFT/LoRA** with 4-bit AWQ quantization, boosting classification accuracy from **60% to 88.4%**
- Optimized inference with vLLM engine, achieving **sub-120ms latency per frame** on NVIDIA Jetson / T4
- **Techstack:** PyTorch, Hugging Face, PEFT/LoRA, vLLM, AWQ Quantization, OpenCV, Docker

Serverless Advanced Hybrid RAG on AWS [↗](#)

Production MLOps, Neo4j + Pinecone, DeepEval

- Architected advanced hybrid retrieval pipeline combining hypothetical document query rewriting (HyDE), dense vector search (Pinecone), and multi-hop knowledge graph traversal (Neo4j)
- Implemented **Reciprocal Rank Fusion (RRF)** followed by a cross-encoder re-ranker (bge-reranker-large), lifting **Recall@5 from 71.3% to 89.6%** and NDCG@10 to 0.86
- Integrated automated CI/CD evaluation test suite using DeepEval (LLM-as-a-Judge), tracking and enforcing **>92% Faithfulness** and **>88% Answer Relevancy** thresholds prior to deployment
- Built parallelized chunking & embedding ingestion pipeline using AWS Lambda and S3, sustaining **sub-250ms end-to-end retrieval latency** at **<\$1.20/month** operational cost
- **Techstack:** Terraform, AWS (Lambda, S3, ECR), Neo4j Aura, Pinecone, Cross-Encoder Reranker, DeepEval, Docker

Certifications

AWS Certified Solution Architect Associate (AWS-SAA-C03) [↗](#)